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Quantifying expert speech: A comprehensive analysis of instructional discourses

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ABSTRACT

Oral communication is a crucial skill in modern society. Nevertheless, it requires sustained practice and constructive feedback. Consequently, several studies have explored the development of oral communication trainers powered by Artificial Intelligence (AI). However, what characterizes expert speech remains unclear, especially given the need to adapt speech to contextual factors. In instructional environments, the speaker's communication proficiency is a key determinant of audience learning outcomes. For this reason, we have analyzed 1250 speeches from five types of instructional discourses: in-person college classes (*Lectures*), online learning lessons (*Online Courses*), instructional animations (*Animated Lessons*), supplementary materials for school and high school (*Supplementary Lessons*), and public presentations (*Public Talks*). We extracted 16 speech metrics, including six additional multiple-participant metrics for *Lectures*. We obtained 250 videos of each discourse type, ensuring a minimum length of five minutes. Our analysis revealed expert values for each speech metric and showed how speech metrics vary across discourse types. We also developed an AI speech classifier that achieved an F1 score of 0.78. The model struggled to identify *Online Courses*, which is consistent with the Uniform Manifold Approximation and Projection analysis, showing that *Online Courses* are closely interjected with the speech of other instructional discourses. Furthermore, we identified distinct speech profiles in *Lectures*, *Public Talks*, and *Online Courses*, highlighting variations in speaking styles. This research provides valuable insights into expert speech in instructional discourses by offering reference values that can help speakers refine their delivery and support researchers in developing more effective speech training systems.

1. Introduction

Oral communication is a key skill in modern society, playing a crucial role in professional success (Coletti et al., 2023; Robles-Bykbaev et al., 2015). As workplace demands evolve, employees are expected to articulate their ideas clearly and effectively to diverse audiences (Albaladejo-González et al., 2024; Stephenson et al., 2025). Oral communication proficiency demands sustained practice and constructive feedback (Haut et al., 2023; van Ginkel et al., 2015). Since most training contexts cannot involve experts in every session, researchers have focused on developing automatic oral communication trainers (Hummel et al., 2025).

To evaluate and provide feedback on oral communication, researchers must understand the characteristics of expert speech (Ochoa, 2022). Researchers need a reference against which to compare the speeches. Characterizing expert speech requires extensive research from a wide range of experts, a sample with few videos and speakers may fail to capture the full variability that exists among different experts and their styles (Domínguez et al., 2024). Moreover, speech varies across discourse types, requiring adaptation to contextual factors such as the audience (Seroka, 2020). For instance, a speaking rate appropriate for a college lecture may not suit a classroom of young children. Consequently, the speech of experts used to compare other speeches should be from the same discourse type and context of application. In addition, speech analysis requires sophisticated processing and must integrate both audio and textual information derived from transcripts (Rasipuram & Jayagopi, 2018). Furthermore, the analysis should employ explainable metrics, enabling trainers and learners to interpret the results effectively and transform them into actionable feedback (Monsalves et al., 2026; Pardo et al., 2025).

In instructional environments, the way in which a speaker delivers information plays a crucial role due to its influence on the audience's learning outcomes (Fyfield, 2021; Tang et al., 2023). Neverthe-

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less, as far as we know, there is no extensive speech analysis of the multiple types of instructional discourses, each with its own unique characteristics. First, the study could report the reference values of speech metrics consistently observed among experts in each type of instructional discourse. In addition, an in-depth analysis can determine what changes and what remains stable in speech across the different types of instructional discourses, defining the speech of each discourse type. Within the analyzed types of discourses, we can also seek potential expert profiles. Therefore, such a study will provide substantial findings into speech in instructional discourses, contributing to advances in this field. Instructors could use the results of this research to compare their own speech with that of experts (Ochoa & Dominguez, 2020). Future researchers could also employ it for the development of instructional speech trainers, incorporating personalized feedback by comparing a user's speech metrics with those of experts (Albaladejo-González et al., 2024; Ochoa, 2022).

Due to the aforementioned contributions, we conducted an extensive study on 1250 speeches from the main five types of instructional discourses (Chorianopoulos, 2018; Gomez et al., 2022; Köse et al., 2021): in-person lectures (*Lectures*), lessons from online learning courses (*Online Courses*), educational animations (*Animated Lessons*), supporting educational videos for schools and high schools (*Supplementary Lessons*), and short public presentations (*Public Talks*). For these videos, we calculated a total of 16 speech metrics for all discourses. Additionally, for lectures, where students could interrupt with questions, we also calculated six multiple-participant speech metrics. We utilized this extensive dataset to answer the following Research Questions (RQs):

- **RQ1**. What defines the speech patterns of each type of instructional discourse?
- **RQ2**. Can speech metrics be used to differentiate types of instructional discourses?
- **RQ3**. Are there different speech profiles in the same type of instructional discourse?

The remainder of this paper is organized as follows. Section 2 provides a background of speech analysis. Section 3 includes the methodology followed to answer the established RQs. Subsequently, Section 4 shows the results obtained, whereas Section 5 elaborates on their discussion. Finally, Section 6 presents the conclusions and future works.

2. Related works

Encouraged by the importance of effective oral communication in contemporary society, several authors have explored its training and evaluation. For the analysis of oral communications, there are two main categories of features to consider: verbal and non-verbal features. Verbal features are speech-related metrics calculated from the audio signal and its transcription (Bartyzel et al., 2025; García et al., 2024; Rasipuram & Jayagopi, 2018). On the other hand, non-verbal features are metrics focused on the speaker's physical behavior, such as gestures, eye contact with the audience, or movement during delivery (Ding, 2024; Domínguez et al., 2024).

Focused on analyzing oral communication, Pereira et al. (2021) examined non-verbal signals in 34 media interviews. A sociometric badge, a wearable electronic device capable of automatically measuring the amount of face-to-face interaction, extracted a total of 21 metrics from each interview. Using Naive Bayes, the authors classified the rating of the communication skills during the interviews with a balanced accuracy of 0.78. Similarly, Rasipuram and Jayagopi (2018) analyzed the communication skills of 100 participants in two interview scenarios: interface-based and face-to-face interviews. The authors extracted verbal and non-verbal metrics and trained several Artificial Intelligence (AI) models. They found that participants were perceived as better communicators in face-to-face settings. Tanaka et al. (2023) created a social skill trainer using a conversational agent. The authors verified the effectiveness of their proposal with 18 participants who rehearsed the tasks of declining requests, listening to others, making requests, and expressing positive feelings. Our work focuses on speech metrics that are independent of the topic and duration, which allows for more reliable comparisons across different speeches. Furthermore, our analysis encompasses a wide range of speech metrics, with 16 global metrics and 6 additional ones specifically for multi-participant settings.

For healthcare settings, communication with patients plays a critical role in maximizing healthcare outcomes (Butow & Hoque, 2020). This communication is complex, as physicians must consider contextual factors and patients' specific needs (Ogbeyemi et al., 2025). Consequently, Kobayashi et al. (2023) used AI to evaluate the communication skills of physicians caring for geriatric patients before and after a comprehensive care communication skills training program. The quality of communication was assessed based on time spent in eye contact, physical touch, and verbal expression. Similarly, Haut et al. (2023) developed SOPHIE, an automated system designed to train doctor–patient communication skills. The system simulates a conversation with a patient seeking information about their prognosis. After the simulation, participants receive detailed feedback, including a transcription of the dialogue with turn-taking analysis, types of questions asked, use of open-ended questions, sentiment trajectory, empathy-related word clouds, personal pronoun usage, speaking rate, readability grade, and hedge word frequency. Moreover, Heallo (Zhang et al., 2022) provides simulated scenarios for healthcare communication tasks. Using calculations and rules defined by experts, it provides interpretable feedback to the user. In addition, communication analysis is also utilized to identify and monitor diseases (Pulido et al., 2020). Alonso Hernández et al. (2022) developed an automatic interviewer that achieved a similar performance to human interviewers for speech Alzheimer's assessment. Another relevant example is (Goyal et al., 2021), where the authors applied speech analysis techniques to diagnose Parkinson's disease. This approach facilitates scalable disease monitoring, providing more objective and consistent assessments over time.

Presentations are an important type of communication that produces high levels of anxiety for inexperienced speakers (Bartyzel et al., 2025; Monteiro et al., 2023). To address this issue, several researchers have analyzed presentations, focusing mainly on the development of training systems. For this reason, (Ochoa & Dominguez, 2020) developed and validated a presentation trainer. The system computed presentation metrics from video, audio, and slides. The feedback was provided by comparing the speaker's metric values with threshold values defined by experts, although these thresholds were not detailed. In subsequent work, the authors also analyzed the verbal and non-verbal presentation metrics of 2000 university students. Similarly, Schneider et al. (2019) developed a presentation trainer with virtual reality. Nevertheless, the authors did not fully disclose and justify the referenced values utilized for the feedback. Tun et al. (2023) stand out for using transfer learning to evaluate presentations, being the source domain videos from interviews. The discourse types analyzed in this manuscript include presentations similar to those examined by the aforementioned works in the *Public Talks* category. In our analyses, we examined how experts communicate in these presentations and identified different expert profiles.

In instructional environments, other researchers have also analyzed the speech of educators (Albaladejo-González et al., 2024; Pardo et al., 2025). Some authors have focused on speech analysis during classes to provide automatic feedback to educators about the time spent on classroom activities, such as lectures or answering questions (Blanchard et al., 2016; García et al., 2024; Schlotterbeck et al., 2021). Among these proposals, TeachFX stands out for its comprehensiveness and popularity (Demszky et al., 2025). Using a mobile application, educators can record their sessions, which TeachFX then transcribes and analyzes through a suite of AI-based algorithms. The resulting class report includes a transcription of the educator's and student's interventions, complemented by an analysis of the educator's talk time, wait time, the incidence of longer student contributions, a word cloud, and the educator's use of questions to engage with the classroom. Other authors have explored the identifi-

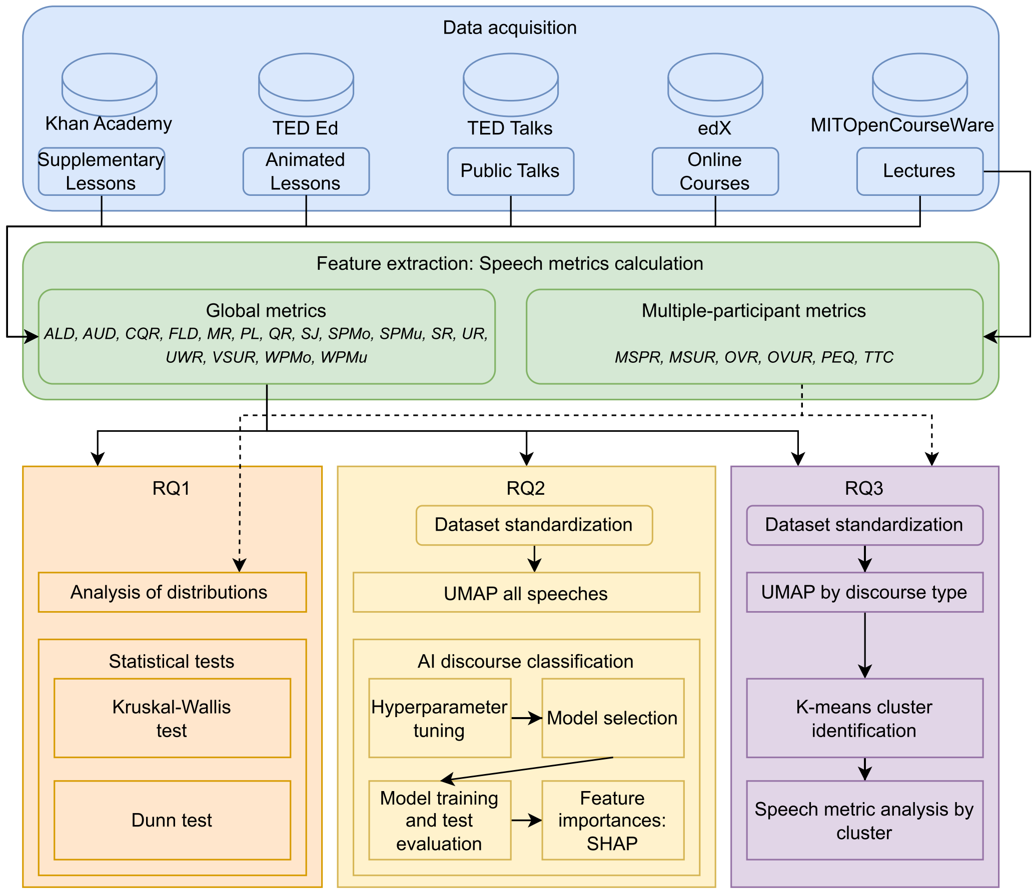

Fig. 1. Summary of the methodology followed to obtain and analyze 1250 videos to answer the RQs.

cation of emotions expressed by educators (Albaladejo-González et al., 2025; Alkhamali et al., 2024; Jie et al., 2020). Finally, we have also found a few works focused on analyzing instructional skills. Tang et al. (2023) implemented an AI-based teacher skills evaluator that measures instructional design, pronunciation, logical statement-making, body language, emotional expression, and teacher–student interaction, among others. Similarly, Jensen et al. (2020) extracted speech metrics from classrooms to evaluate teacher discourse based on the use of instructional talk, questions, authentic questions, student evaluation, cognitive level, use of disciplinary terms, the explanation process, and the goals of an activity. Our work once again stands out for the range of speech metrics analyzed and for covering multiple discourse types within the instructional context.

Despite the existence of several studies on communication analysis, we have identified a lack of clear guidance on what defines expert speech. To use reference values as benchmarks for speech assessment, it is necessary to have an adequate validation and justification (Albaladejo-González et al., 2024; Ochoa & Dominguez, 2020). Given its importance in instructional environments, we have addressed this problem by analyzing the speech of experts in 1250 instructional videos across five different formats. We aim for our experiments' results to serve as a practical benchmark to compare observed speeches and also enable interpretable feedback.

3. Methodology

In this section, we introduce the methodology followed in our experiments to answer the aforementioned RQs. Fig. 1 depicts this methodology, including the data acquisition, feature extraction, and the analyses conducted for each RQ. The experiments presented in this manuscript were approved by the Research Ethics Committee of the University of Murcia, with the identifier M10/2025/104 on March 7, 2025.

3.1. Data acquisition

Our first step was to gather videos from various types of instructional discourses. We focused on speeches in English because of its large number of speakers and its frequent use in educational and academic contexts. We aimed to include the main instructional discourse types, prioritizing the diversity of characteristics among them and their open availability (Chorianopoulos, 2018; Köse et al., 2021). For each type, we selected sources where the speakers' communication skills and knowledge about the discussed topic were widely recognized. The selected discourse types and their sources are described below:

- ***Lectures***: This category encompasses face-to-face college lectures. We downloaded 250 lectures from MITOpenCourseWare and manually filtered the videos to select only in-person MIT lectures from 106 different courses. This type of discourse is long, with an average speech duration of 54 minutes and 29 seconds, and despite being rehearsed, the preparation is low due to its length. Additionally, these are live speeches where the speaker cannot edit the video or repeat segments at will. It is also worth noting their technical and specialized nature, as they are intended for a specific college-level audience.
- ***Online Courses***: In digital learning courses, instructors usually rely on instructional videos to deliver content. We collected 250 videos from edX, corresponding to 92 courses offered by 24 different institutions. The average speech duration is 7 minutes and 47 seconds. These discourses are pre-recorded, allowing educators to edit the video, insert visual aids, or repeat explanations to improve clar-

Table 1
Description of the types of discourses analyzed.

	Source	Average speech duration	Technical	Instructional purpose	Live	Audience	Visual support
Lectures	MITOpen-CourseWare	3270s	Yes	High	Yes	College students	Slides and/or blackboard
Online Courses	edX	467s	Mix	High	No	Mix	Mix
Animated Lessons	TED Ed	288s	No	Medium	No	Broad audience	Animations
Supplementary Lessons	Khan Academy	474s	No	Complement education	No	School and high school students	Digital blackboard
Public Talks	TED Talks	707s	No	Low	Yes	Broad audience	Slides

ity. It is also worth noting that the targeted audience changes depending on the course topic. A wide variety of topics is represented, ranging from technical subjects, such as mathematics, to more general themes, for instance, basic economics.

- ***Animated Lessons***: In this category, we included instructional videos that use animated illustrations to convey concepts. We downloaded 250 animations on different topics from TED Ed. These videos are short, with an average speech duration of 4 minutes and 47 seconds, and are highly edited. The content is scripted and carefully refined. These videos are not live, and their technical depth is limited, focusing instead on making content accessible and engaging. The target audience is broad, often including learners of all ages, and the tone is typically general and explanatory.
- ***Supplementary Lessons***: This category integrates videos to complement school and high school education. For this purpose, we collected 250 videos from Khan Academy. These videos are concise, with an average speech duration of 7 minutes and 54 seconds, and are typically pre-recorded in a digital blackboard format. The speaker frequently writes and draws on the digital blackboard while explaining concepts.
- ***Public Talks***: This category contains live presentations delivered in front of an audience. We obtained 250 videos from TED Talks. These presentations are moderately long, with an average speech duration of 11 minutes and 46 seconds. Since they are delivered live, there is no possibility of editing or repeating segments. The content is usually polished and rehearsed, but not overly technical. The target audience is broad, and the talks encompass a diverse range of topics, from personal experiences to scientific research. These presentations are usually accompanied by slides.

The first four types of discourses primarily belong to the instructional field, where effective speech is crucial for fostering a meaningful learning experience. In the case of *Public Talks*, we analyzed TED Talks presentations, which are internationally renowned. While these talks also pursue an instructional purpose, their primary aim is the dissemination of ideas across diverse domains. Table 1 summarizes the main characteristics of each type of discourse and its respective source.

For the 1250 discourses, we selected only videos with a length of five minutes or longer. This threshold ensured a sufficiently large audio sample for analysis and reduced the impact of short outliers. However, it is worth noting that the length of the speech could be shorter than five minutes due to silences and pauses in the recordings. Videos from courses on language learning, public speaking, or communication were excluded, as instructors in these subjects often used highly varied speech, frequently incorporating deliberate exaggerations and contrasting examples of effective and ineffective communication practices. In *Lectures*, we removed online lectures from MITOpenCourseWare that were recorded during the COVID-19 lockdown. These discourses were not in-person and also showed a lack of experience with the format; therefore, we could not use them for the *Lectures* or *Online Courses* discourses categories. For *Online Courses*, we conducted an extensive manual search on edX, as many courses do not allow video downloads and have a shorter duration of five minutes. From the available courses in *Lectures* and *Online Courses*, we randomly selected a maximum of three videos, aiming to provide a more diverse sample. For *Supplementary Lessons*, we downloaded Khan Academy videos from YouTube because the platform loads the videos from Khan Academy's channel. The videos were organized by playlist, and we only downloaded a video from each playlist to cover as many topics as possible. Despite all these filters, we were able to obtain 250 videos for each type of discourse, generating a balanced dataset of 1250 discourses.

3.2. Feature extraction

Once we obtained the 1250 discourses, we calculated 16 speech metrics for all of them. Given the absence of body recordings in several collected discourses, we focused on speech as a modality consistently available across the dataset. These metrics were designed to characterize overall speech delivery and were defined as duration-independent. Otherwise, the three RQs could yield misleading results by distinguishing discourses primarily based on their duration. For this reason, some metrics were computed using 60-second time windows and then averaged to ensure duration independence. Moreover, the speech metrics should be topic-agnostic. For instance, they should not differ between a computer science and a history lecture solely due to subject matter. Consequently, they provide a more robust benchmark for future researchers comparing speech samples across contexts. In discourses with multiple speakers, which in our case only occurred in *Lectures* that could include brief student interventions, the speech metrics were calculated for the main speaker, defined as the person who speaks the most. It is also important to clarify that an intervention or speech can be divided into different utterances based on the presence of a significant pause or silence. These 16 speech metrics are defined below:

- **Average Lapse Duration (ALD)**: Total silence duration divided by the number of silences. This metric is calculated by each window, and then we return an average value.
- **Average Utterance Duration (AUD)**: The average duration of the utterances of the speaker in seconds.
- **Confused Questions Ratio (CQR)**: Number of confused questions (questions of one word, such as ok?) divided by the word count.
- **Filler Lexical Density (FLD)**: Number of filler words divided by the word count.
- **Mumble Ratio (MR)**: This metric evaluates the extent of speech that cannot be attributed to a particular speaker during the diarization process. These instances could be from the main speaker, but the voice activity detection cannot associate them with the main speaker or any other individuals.
- **PoLarity (PL)**: Emotional polarity of the speech is calculated by TextBlob (Aljedaani et al., 2022). The polarity score is a float within the range [-1.0, 1.0]. A value close to -1 indicates negativity, 0 neutrality, and 1 positivity.
- **Questions Ratio (QR)**: Number of questions divided by the word count. This ratio includes questions from the instructor to the audience.
- **SubJectivity (SJ)**: Subjectivity of the speech calculated by TextBlob (Aljedaani et al., 2022). A value close to 0 indicates objectivity, and a value close to 1 indicates subjectivity.
- **Syllable Per Minute overall (SPMo)**: Total number of syllables in the entire speech divided by the total speech duration in minutes, including silences.

- **Syllable Per Minute by utterance (SPMu)**: Total number of syllables in the entire speech divided by the total speech duration of the utterances in minutes (speaking time).
- **Silence Ratio (SR)**: Total duration of silences divided by the total duration.
- **Utterances Ratio (UR)**: Calculation of the number of utterances of the speaker averaged over windows.
- **Unique Words Ratio (UWR)**: Number of unique words divided by the total number of words.
- **Very Short Utterances Ratio (VSUR)**: Number of short utterances (less than 2 seconds) divided by the total.
- **Words Per Minute overall (WPMo)**: Total number of words in the entire speech divided by the total speech duration in minutes, including silences.
- **Words Per Minute by utterance (WPMu)**: Total number of words in the entire speech divided by the total speech duration of the utterances in minutes (speaking time).

Furthermore, we calculated six multiple-participant speech metrics. In our dataset, these metrics were computed exclusively for *Lectures*, as this was the only discourse type that involved multiple participants. In *Lectures*, student interventions mainly involved answering the teacher's questions or asking questions. Although audience questions may occur in *Public Talks*, our data collection did not include them. The six multi-participant speech metrics are defined as follows:

- **Main Speaker Participation Ratio (MSPR)**. The ratio of participation by the main speaker, the teacher, during the window, averaged across all windows.
- **Main Speaker Utterances Ratio (MSUR)**. The ratio of utterances of the main speaker during each window, averaged across all windows.
- **Overlapping Rate (OVR)**. The rate of overlapping utterances of different speakers, averaged across all windows.
- **Overlapping Utterances Rate (OVUR)**. The ratio of utterances that overlap, averaged across all windows.
- **Participation Equality (PEQ)**. An indicator that assesses the balance of participation among different speakers, averaged across all windows. It is calculated following the methodology outlined in (Lai et al., 2013). Values close to 1 indicate an equal distribution of participation.
- **Turn Taking Count (TTC)**. The number of turn changes, averaged across all windows.

To calculate these speech metrics, it was necessary to obtain the transcript of the speech. For this task, we used Whisper (Radford et al., 2022), a well-known automatic speech recognition model based on transformer architecture. Specifically, we employed the large version of Whisper, which has 1550 million parameters. To estimate PL and SJ, we employed TextBlob (Aljedaani et al., 2022). In addition, for the audio analysis, we also used the Python library Librosa (McFee et al., 2015). We utilized Pyannote for the diarization process (Plaquet & Bredin, 2023). We also employed other common Python libraries and packages, such as Numpy and Pandas.

3.3. Experimental methods

RQ1 examined how speech metrics vary across the different discourse types. First, we computed and compared the mean, standard deviation, and the quartiles (Q1, Q2, Q3) of each metric for each discourse type. In addition, we applied the Kruskal–Wallis test to determine whether these metrics showed significant differences depending on the discourse type. The Kruskal–Wallis test was chosen because the data did not meet the assumptions of normality and homogeneity of variances required for the ANOVA test. Additionally, to locate the differences, we used Dunn's test with Bonferroni correction.

To address RQ2, we aimed to assess how well speeches could be separated according to their discourse type. For this purpose, we used Uniform Manifold Approximation and Projection (UMAP), a dimensionality reduction technique commonly used for data visualization. For the UMAP representation, we applied a standardization of the dataset, scaling the speech metrics to a normal distribution with mean 0 and standard deviation 1. This process aimed to minimize the influence of differences in the speech metrics scales, ensuring that all metrics contributed equally to the UMAP calculations. UMAP results were displayed through a scatter plot, assigning a different color to the inputs based on their discourse types.

We also analyzed this separability by training and evaluating an AI model to classify the discourses according to their type. For this experiment, we split the dataset into 70% for training and 30% for test. We added the condition that videos from the same course had to be placed in the same subset, and we attempted to keep both datasets as balanced as possible. Ultimately, from the 1250 videos, 75 examples from each class were included in the test set. On the training set, we performed a grid search for hyperparameter tuning using 10-fold cross-validation. During this grid search, all configurations were tested using both the standardized dataset and the unscaled dataset. In the cross-validation process, we also ensured that videos from the same course were always included in the same fold. The models configured and evaluated during cross-validation included were Random Forest (RF), Adaptive Boosting (AdaBoost), K-Nearest Neighbors (K-NN), Stochastic Gradient Descent Classifier (SGD), Support Vector Classifier (SVC), Decision Tree (DT), and MultiLayer Perceptron (MLP). This selection covered bagging and boosting models, simple classifiers, and neural networks. For the aforementioned models, we utilized the implementations available in Scikit-learn (Pedregosa et al., 2011). Then, the model with its configuration that achieved the best performance was trained on the full training set and evaluated on the test set. We also used explainable AI techniques, specifically SHapley Additive exPlanations (SHAP), to analyze the importance that the best model assigned to each speech metric. SHAP is a post-hoc explainability method based on cooperative game theory that assigns an importance value to each feature. This technique is widely utilized to understand which input features an AI model is considering when providing a prediction. Features with higher importance values are significant for the prediction, while those with lower values are less influential. In our research, SHAP allowed us to observe which speech metrics were most relevant for the AI model in differentiating between the various types of discourses. To calculate SHAP, we utilized the Python library Explainerdashboard (Oege Dijk et al., 2025).

The final analyses aimed to identify distinct groups of speeches within each discourse type, addressing RQ3. We identified speech profiles only for *Online Courses*, *Lectures*, and *Public Talks*, since the variety of speakers in *Animated Lessons* and *Supplementary Lessons* was limited. Furthermore, these videos did not show the speakers, and therefore, we could not verify whether the groups were based on speaker characteristics such as gender or age. In addition, these videos were highly predefined by their institutions, and because they were pre-recorded, the variability was expected to be minimal. For *Lecture* videos, we also employed the six multiple-participant metrics, which were exclusively available for this type of discourse. First, we standardized the speech metrics, applied UMAP, and then we utilized K-means to identify the clusters. To determine the optimal number of clusters, we calculated the Within-Cluster Sum of Squares (WCSS) for different values of k and selected the value beyond which further increases yielded only marginal improvements. We also tested other dimensionality reduction algorithms, including Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE). In addition, to identify the clusters, we also tested Density-Based Spatial Clustering of Applications with Noise (DBSCAN), Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN), and Gaussian Mixture Models. However, in a scatter plot representation of the clusters, the combination of UMAP and K-means demonstrated effective cluster identification. The clusters were also highly interpretable based on

Table 2
Summary of the significant differences among the types of discourses (RQ1).

Comparison	Total	Metrics
Public Talks vs *Supplementary Lessons*	11	ALD, AUD, FLD, MR, PL, SJ, SR, UR, UWR, WPMo, WPMu
Online Courses vs *Supplementary Lessons*	10	ALD, AUD, PL, QR, SJ, SR, UWR, VSUR, WPMo, WPMu
Online Courses vs *Public Talks*	9	ALD, MR, PL, QR, SJ, SR, UR, VSUR, WPMo
Lectures vs *Supplementary Lessons*	14	ALD, AUD, CQR, FLD, MR, QR, SJ, SPMo, SPMu, SR, UR, UWR, VSUR, WPMo
Lectures vs *Public Talks*	13	ALD, CQR, FLD, PL, QR, SJ, SPMo, SPMu, SR, UWR, VSUR, WPMo, WPMu
Lectures vs *Online Courses*	13	ALD, AUD, CQR, FLD, MR, QR, SPMo, SPMu, SR, UR, UWR, VSUR, WPMu
Animated Lessons vs *Supplementary Lessons*	11	ALD, FLD, QR, SJ, SPMu, SR, UR, UWR, VSUR, WPMo, WPMu
Animated Lessons vs *Public Talks*	14	ALD, AUD, CQR, MR, PL, QR, SPMo, SPMu, SR, UR, UWR, VSUR, WPMo, WPMu
Animated Lessons vs *Online Courses*	10	AUD, SJ, SPMo, SPMu, SR, UR, UWR, VSUR, WPMo, WPMu
Animated Lessons vs *Lectures*	14	ALD, AUD, CQR, FLD, MR, QR, SJ, SPMu, SR, UR, UWR, VSUR, WPMo, WPMu

the speech metrics, and K-means is a simpler and more explainable algorithm.

4. Results

This section contains the results of the experiments conducted to answer each RQ.

4.1. RQ1. What defines the speech patterns of each type of instructional discourse?

Fig. 2 provides an overview of the distribution of each speech metric across the various types of discourses. To enhance interpretability, all outliers were removed. Similarly, Fig. 3 shows the distribution of the speech metrics, which were calculated only for multiple participants' discourses in our case, *Lectures*, due to the students' interventions. In addition, Table A.4 offers a comprehensive summary of the dataset without outlier removal, reporting the first, second, and third quartiles, along with the mean and standard deviation for each speech metric and discourse type. To support their analysis, we employed the Kruskal–Wallis test to assess whether statistically significant differences existed in each metric across discourse types. Subsequently, the Dunn test was applied to identify pairwise differences, with the results also reported in Table 2.

The patterns observed in Fig. 2 indicate that the metrics ALD and SR successfully differentiate between most discourse types. According to Table 2, SR exhibits significant differences across all discourse categories, whereas ALD differentiates among nearly all categories except for *Animated Lessons* and *Online Courses*. These findings suggest that the time dedicated to silence varies depending on the discourse format. *Lectures* show more time dedicated to silences, which may be attributed to their extended length, they are not recorded, and a lack of post-production editing.

The analysis of AUD reveals that *Animated Lessons* and *Supplementary Lessons* tend to include longer utterances. VSUR shows differences across the discourse types based on the proportion of very short utterances. This proportion is significantly lower in *Animated Lessons* and higher in *Lectures*. VSUR proves useful in distinguishing between discourse formats, with the exception of *Public Talks* and *Supplementary Lessons*, where it does not reveal significant differences. In contrast, the UR is higher in *Public Talks* and *Lectures*. *Animated Lessons* combine long utterances with a relatively low number of utterances. *Supplementary Lessons* exhibit the same phenomenon, but with less pronounced effects in UR due to their faster speech speed, which increases the number of utterances.

To analyze the speech speed, we calculated four speech metrics. The words per minute metric, when calculated across the entire discourse including periods of silence (WPMo), reveals significant differences between most discourse types, except between *Lectures* and *Online Courses*. In contrast, the syllables per minute metric, calculated using the same approach (SPMo), does not clearly distinguish the discourse types. Both measures reflect the overall speaking rate, though the syllable-based metric is also influenced by word length. When calculated based solely on speaking time, both metrics yield higher values but reveal few relevant changes (WPMu and SPMu). Considering all four variants of speech speed, we appreciate that *Animated Lessons* have the slowest speech speed, which may reflect their focus on accessibility, non-technical content, and the possibility of editing due to their recorded format.

Regarding emotional content, PL across discourses tend to lean toward neutrality or positivity, with *Public Talks* exhibiting the most positive tone. In terms of SJ, *Public Talks* and *Animated Lessons* exhibit the highest levels of subjectivity. These results suggest that *Public Talks* are generally more emotionally expressive, possibly due to the inclusion of personal narratives and experiences.

The metric UWR exhibits considerable variation among the five discourse types. *Lectures* display the lowest proportion of unique words, possibly due to their technical content, duration, and live nature; the speech in them tends to be more repetitive or less varied. In contrast, *Online Courses* exhibit a wide range of values, potentially influenced by differences in course content and audience. *Animated Lessons* show the highest proportion of unique words, marking a distinct contrast with the other categories. The statistical analysis summarized in Table 2 indicates that UWR reveals significant differences across all categories except between *Online Courses* and *Public Talks*. The FLD metric, which captures the proportion of filler words, remains consistently low across all discourse types, as the recordings involve expert speakers. Nonetheless, slightly higher values are observed in *Lectures*. The MR metric, which quantifies the presence of mumbles, presents higher values in *Public Talks* and *Lectures* compared to *Animated Lessons*, *Supplementary Lessons*, and *Online Courses*. This may be partially explained by the fact that the former are delivered live with an audience, while the latter are prerecorded and typically subject to editing. It is important to note that, in a few cases, mumbled utterances may originate from the audience rather than from the primary speaker.

In relation to the presence of questions, measured by the CQR and QR metrics, the highest values are found in *Lectures*. This pattern aligns with the in-person nature of *Lectures*, where it is more common for speakers to pose open-ended questions, even without anticipating a response from the audience. The multiple-participant speech metrics, calculated only for *Lectures*, reveal that the instructor spoke for the majority of the time. Fig. 3 shows that the student participation was low, indicating a predominantly unidirectional classroom style. This finding is relevant for comparison with the other instructional discourses analyzed, as it provides evidence that the selected recordings represent lecture-style classes rather than highly collaborative classroom dynamics.

4.2. RQ2. Can speech metrics be used to differentiate types of instructional discourses?

The experiments in RQ2 aimed to analyze the separability of the different types of discourses based on the speech metrics. Fig. 4 shows a UMAP projection of the five discourse types. In this figure, we observe a strong separation among *Lectures*, *Supplementary Lessons*, and *Animated Lessons*. *Public Talks* are in the center of the figure, but also overlap with other types of discourses. *Online Courses* do not form a clearly identifi-

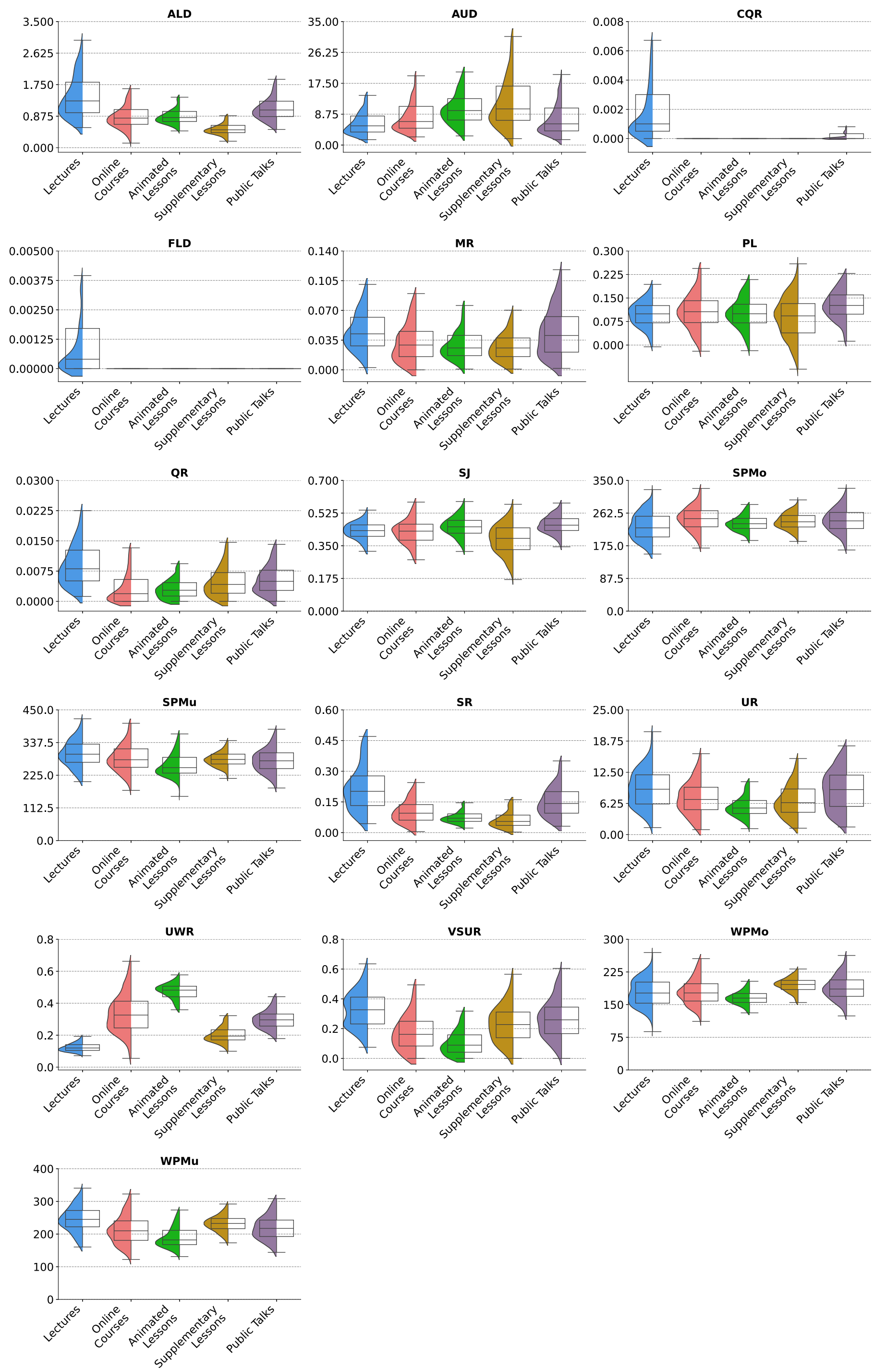

Fig. 2. Summary of the speech metrics for each type of discourse. Outliers removed for clarification (RQ1).

able cluster; instead, they are dispersed around the UMAP projection, overlapping with all other discourse types.

We also evaluated the performance of an AI model in classifying discourse types based on speech metrics. First, we performed hyperparameter tuning. Table 3 presents the performance of the optimal configuration of each model. All models demonstrated equal or improved performance when using the scaled dataset. Among them, the MLP achieved the best results, reaching an accuracy of 0.80 and an F1 score of 0.79. This MLP model has two hidden layers of 50 neurons, utilizes the logistic activation function, an alpha value of 1, without early stopping, a constant learning rate of 0.0001, a maximum of 200 iterations, and the LBFGS solver for weight optimization. Fig. 5 shows the performance of the MLP on the test subset, achieving an accuracy of 0.79 and an F1 score of 0.78. Although the overall performance is acceptable, it struggles to accurately identify *Online Courses* speeches. This is consistent with the UMAP projection shown in Fig. 4, since we identified that *Online Courses* are highly intersected with the clusters of other types of discourses. In contrast, *Lectures* are the discourse type most accurately identified by the MLP. Finally, we used SHAP to analyze metric importance in the MLP model, as shown in Fig. 6. The metric UWR was identified as the most influential, with twice the importance of the next most relevant metric. In RQ1, we observed that this metric shows sig-

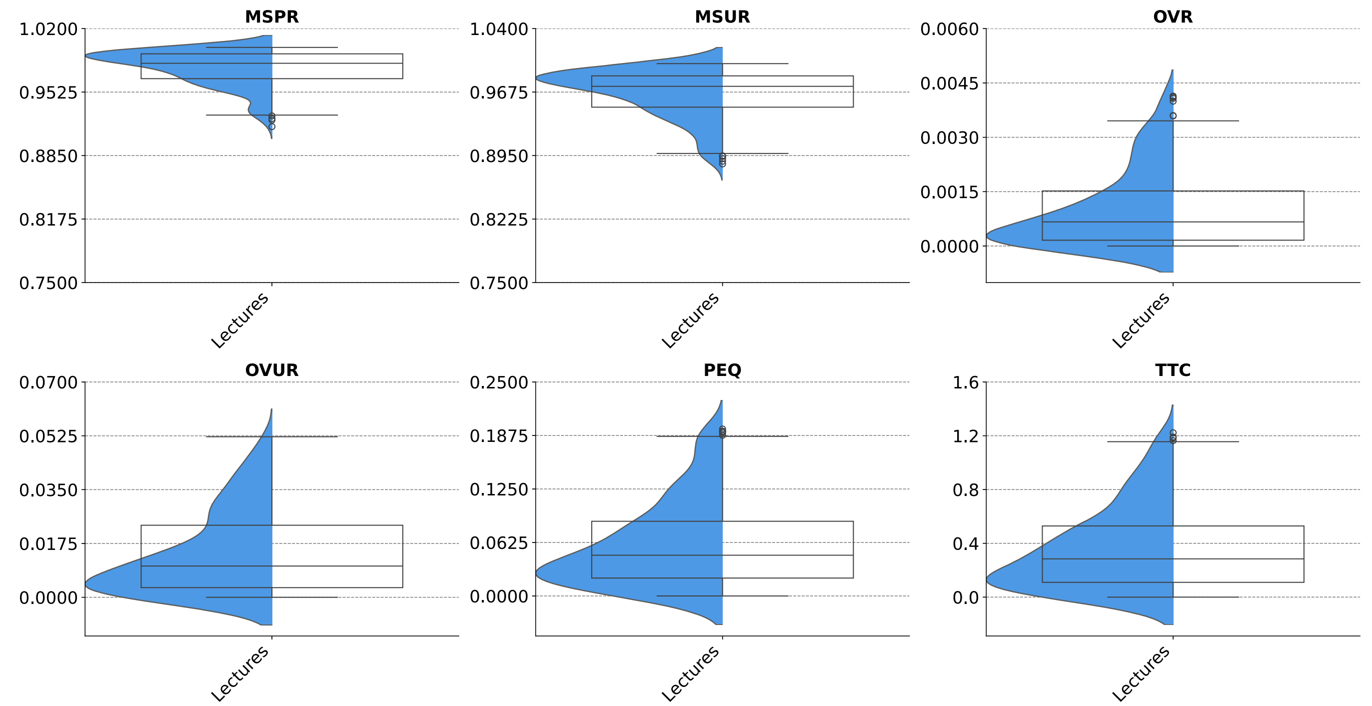

Fig. 3. Summary of the speech metrics only for multiple speakers, in our case, *Lectures* (RQ1).

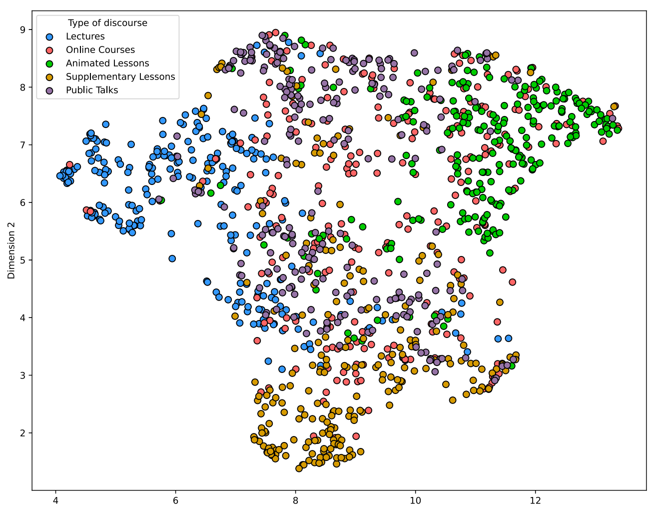

Fig. 4. UMAP scatter plot of all types of discourses (RQ2).

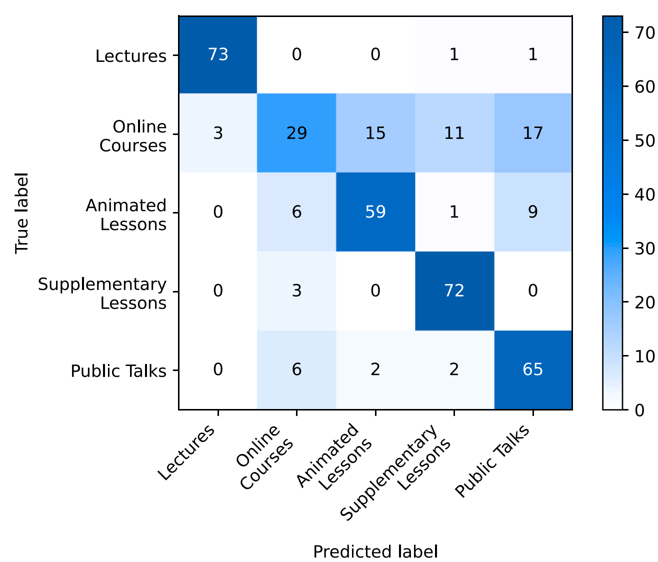

Fig. 5. Confusion matrix of the MLP in the test subset (RQ2).

nificant differences among all discourse types, except between *Online Courses* and *Public Talks*. In contrast, MR was the least relevant metric for the AI model. Additionally, RQ1 revealed that MR only shows differences between live discourses with an audience (*Lectures* and *Public Talks*) and recorded discourses (*Online Courses*, *Animated Lessons*, and *Supplementary Lessons*).

4.3. RQ3. Are there different speech profiles in the same type of instructional discourse?

For this RQ, we aimed to identify different groups of speeches within each discourse type based on speech metrics. We focused on *Lectures*, *Online Courses*, and *Public Talks*, excluding *Supplementary Lessons* and *Animated Lessons* because they follow a highly predefined style, and it is not possible to reliably identify the number of different speakers present in all the videos. First, we standardized the speech metrics, applied UMAP, and then we utilized K-means to identify the clusters. To determine the optimal number of clusters, we calculated the WCSS for different values of k, and selected the value of k beyond which further increases yield only marginal improvements. Fig. 7 displays the UMAP projection with clusters identified by K-means.

K-means identified four profiles of *Lectures*. Fig. 8 shows the metrics that vary across clusters revealed in *Lectures*. Cluster 1 includes speeches with low AUD and high UR and VSUR compared to the other clusters, indicating the use of shorter utterances. This cluster is also characterized by frequent use of questions and mumbles, as reflected in the CQR, QR, and MR metrics. Cluster 2 is characterized by a low use of questions, as evidenced by the lowest values in CQR and QR. It also has the highest UWR, suggesting a greater use of unique words. Cluster 3 stands out for its short pause durations, as measured by both ALD and SR. Additionally,

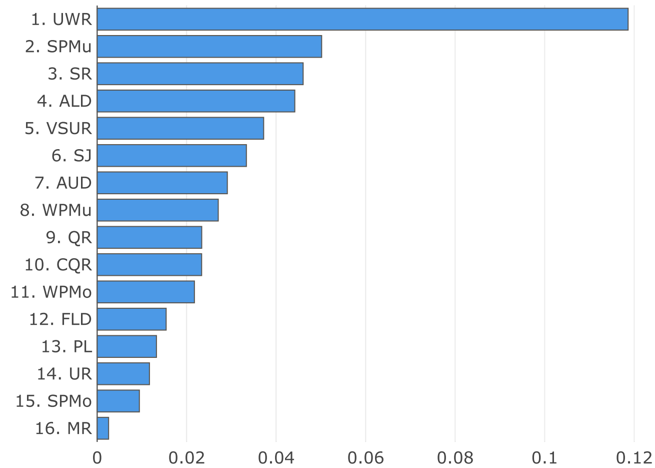

Fig. 6. Importance of each speech metric for the MLP, calculated through SHAP (RQ2).

Table 3
Grid search performances of the best configuration for each AI model (RQ2).

Model	Accuracy	Precision	Recall	F1 score
AdaBoost	0.689	0.680	0.685	0.671
KNN	0.683	0.675	0.681	0.655
MLP	**0.799**	**0.796**	**0.796**	**0.790**
RF	0.791	0.792	0.789	0.782
SGD	0.730	0.730	0.732	0.719
SVC	0.766	0.762	0.767	0.758
Tree	0.731	0.734	0.727	0.713

AUD and VSUR indicate that speeches in this cluster tend to include longer utterances compared to the others. Cluster 4, similar to Cluster 1, uses shorter utterances than Clusters 2 and 3. Cluster 4 is also notable for its lack of emotional expression, with the lowest levels of subjectivity and polarity.

Analyzing *Online Courses*, K-means distinguished three profiles. Fig. 9 shows the metrics that vary across clusters identified in *Online Courses*. Cluster 1 stands out for its shorter utterances, as indicated by the AUD, UR, and VSUR values. It also includes the most frequent use of mumbles. Cluster 2 shows the lowest QR and WPMo, and the highest UWR, suggesting speeches with fewer questions, more unique words, and a slower speaking rate. Cluster 3 is characterized by the highest AUD and WPMo and the lowest UR and VSUR, indicating longer utterances and a faster speech.

Through the analysis of *Public Talks*, K-means revealed four distinct profiles. Fig. 10 illustrates how key speech metrics vary across the resulting clusters. Clusters 1 and 2 are very similar; however, Cluster 1 has the highest frequency of unique words, while Cluster 2 shows the fastest speech speed. In contrast, Cluster 1 has the slowest speech. Cluster 3 also stands out for its longer utterances, as indicated by the AUD, UR, and VSUR values. The same metrics indicate that Cluster 4 includes the shortest utterances.

5. Discussion

This section provides an interpretation of the results obtained, their implications, and limitations.

5.1. Interpretation of the results

Lectures are characterized by their long duration, live recording, and technical content, which is aimed at an academic audience. The analyzed lectures included more time dedicated to silences, shorter utterances, and a higher frequency of mumbles and filler words. The proportion of unique words was generally lower than in the other discourse types, which might reflect a certain degree of repetitiveness associated with the use of technical terms related to the topic. Overall, *Lectures* seemed to present less refined speech patterns, which could be related to their extended duration and live format. The multiple-participant speech metrics suggested a predominantly unidirectional classroom style, with the instructor speaking during most of the session. Among the discourse types, *Lectures* were those better recognized by the MLP. K-means clustering identified four distinct groups of *Lectures*, each differentiated by variations in utterance length, question usage, pause duration, and emotional metrics. One group showed shorter utterances combined with frequent questions and mumbles; another displayed a higher diversity of vocabulary but fewer questions; a third tended to include longer utterances and shorter pauses; and a fourth was characterized by shorter utterances and limited emotional expression.

Public Talks are also live discourses, but highly rehearsed. The analyzed *Public Talks* tended to include relatively long pauses, short utterances, and a noticeable presence of mumbles, although these seemed less frequent than in *Lectures*. Regarding filler words, lexical diversity, and speech speed, their values were similar to those of other discourse types. *Public Talks* also tended to exhibit higher subjectivity and more positive emotional expressions, which might be related to the frequent use of personal narratives and experiences. Overall, some speech metrics placed them closer to *Lectures*, possibly due to the shared live delivery format. However, the contrasts with recorded discourses appeared less pronounced than in *Lectures*, perhaps because *Public Talks* are shorter and more carefully prepared. The K-means analysis suggested four groups within the *Public Talks*, each differentiated by speech rate, utterance length, and lexical richness. Two groups showed broadly similar patterns, although one contained a higher proportion of unique words while the other displayed faster speech. The remaining two groups were characterized by relatively longer or shorter utterances.

In contrast to the live format of *Public Talks* and *Lectures*, *Online Courses* are short and pre-recorded instructional videos. Unlike the more distinctive patterns observed in live discourses, *Online Courses* did not appear to present particularly extreme values in the speech metrics considered. Instead, their values often seemed closer to those of other discourse types. This may suggest that *Online Courses* exhibit relatively average or typical speech patterns among the analyzed discourses. Therefore, the UMAP scatter plot (Fig. 4) reveals that *Online Courses* are largely dispersed and overlap with various discourse types. Consistently, the MLP encountered more difficulty in identifying *Online Courses*.

The K-means analysis revealed three groups of *Online Courses*, distinguished by differences in utterance length, speech speed, lexical diversity, and the presence of questions or mumbles. One group exhibited shorter utterances combined with frequent mumbles, while another displayed fewer questions, higher lexical diversity, and slower speech. A third group tended to longer utterances and faster speech.

Animated Lessons in our dataset were scripted and pre-recorded discourses, designed with substantial visual support. They tended to show relatively slower speech rates and higher proportions of unique words, combined with long utterances and few filler words and mumbles. These results may suggest more refined and quality speech compared to other discourse types. In contrast, the subjectivity was higher compared to other types of discourses.

We observed a different pattern in *Supplementary Lessons*, which are pre-recorded videos, focused on support school and high school education. These discourses dedicated less time to pauses and longer utterances. In addition, they tended to show faster speech rates and lower lexical diversity, although not as much as *Lectures*.

5.2. Implications

The results from RQ1 allowed us to identify key speech differences across various types of instructional discourses. Since the dataset includes speeches delivered by expert speakers, these findings, especially

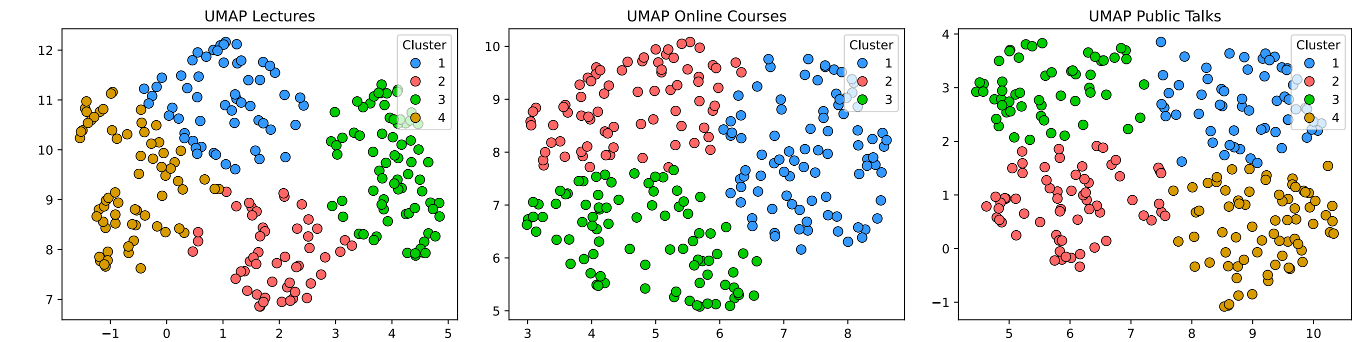

Fig. 7. UMAP scatter plot with K-means clusters (RQ3).

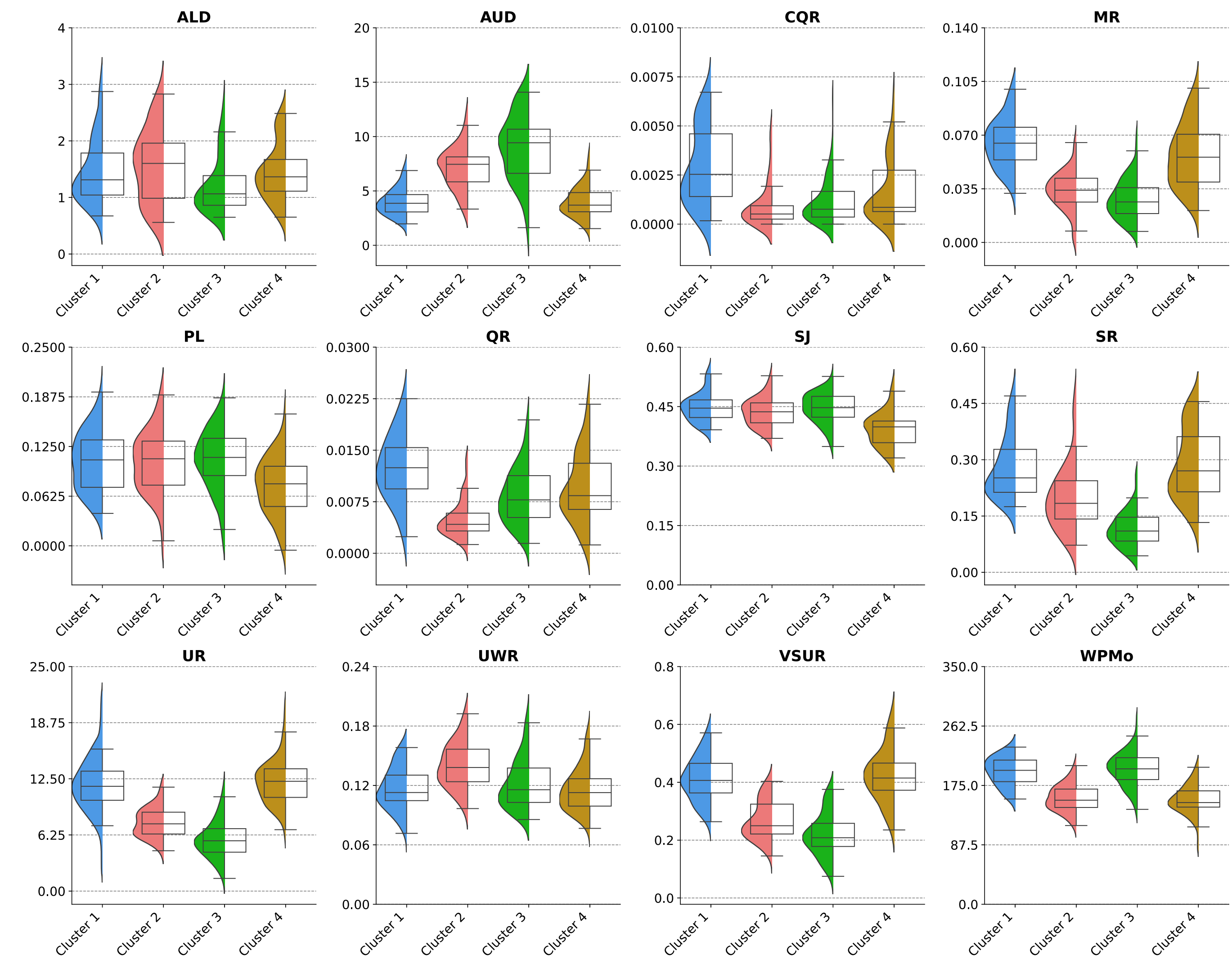

Fig. 8. Summary of profiles identified in *Lectures* (RQ3).

Table A.4, can serve as a reference for evaluating instructional speeches and determining how closely they align with expert speeches. For instance, the speeches from TED Talks can be used as benchmarks to assess similar presentations by inexperienced speakers and to provide them with targeted feedback. This feedback would be interpretable, as it could highlight specific speech metrics that deviate from expert values. For example, a public presentation with a WPMo of 250 is above the third quartile of the experts in *Public Talks*; therefore, the speaker would receive the recommendation to reduce the speaking speed. In addition to quartile-based comparisons, researchers can also apply other approaches. For instance, they can compare a speech against the mean and standard deviation of the expert metrics, which are also reported in the aforementioned table. Moreover, discourse trainers can deliver this feedback in real-time, enabling the speaker to adjust their delivery during the discourse, or afterward as part of a reflective analysis to guide future improvement. In a *Lecture*, real-time feedback allows instructors to identify moments when their speech reaches inadequate levels and modify it to prevent student learning loss during the lesson. Similarly, platforms hosting instructional videos can use these metrics to evaluate uploaded content as part of a filtering process. The different profiles

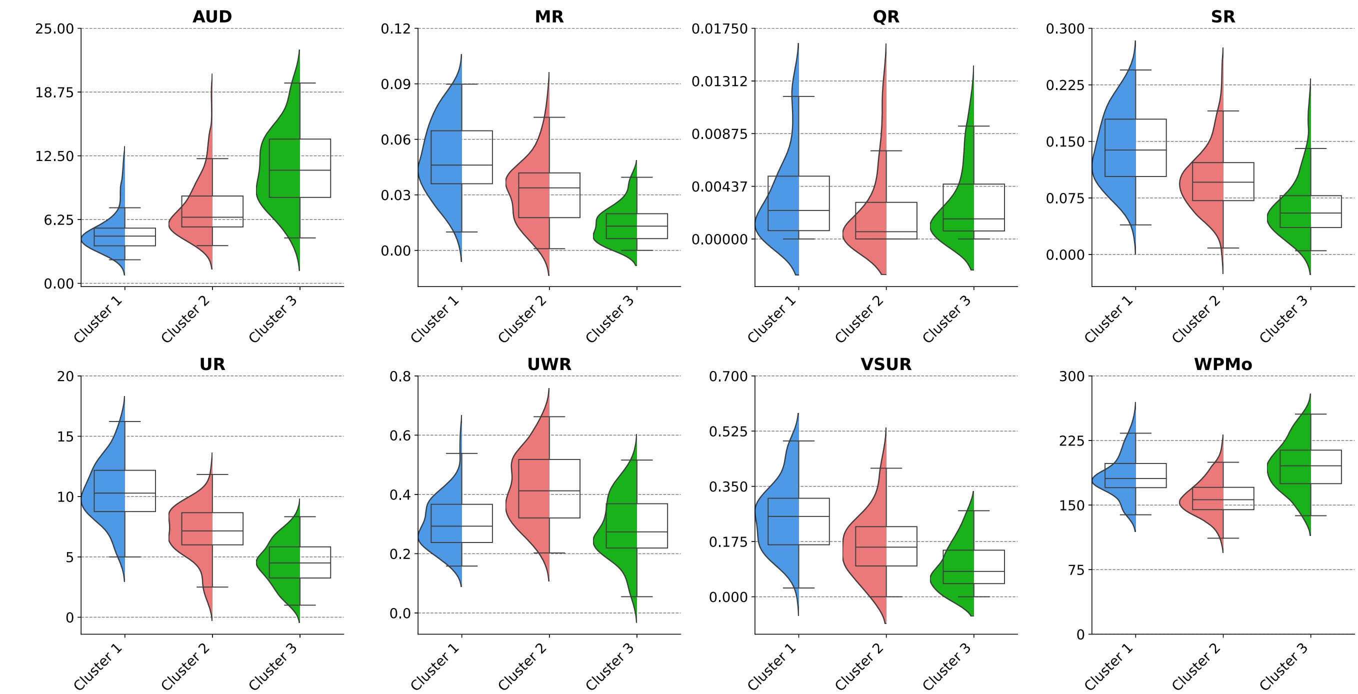

Fig. 9. Summary of profiles identified in *Online Courses* (RQ3).

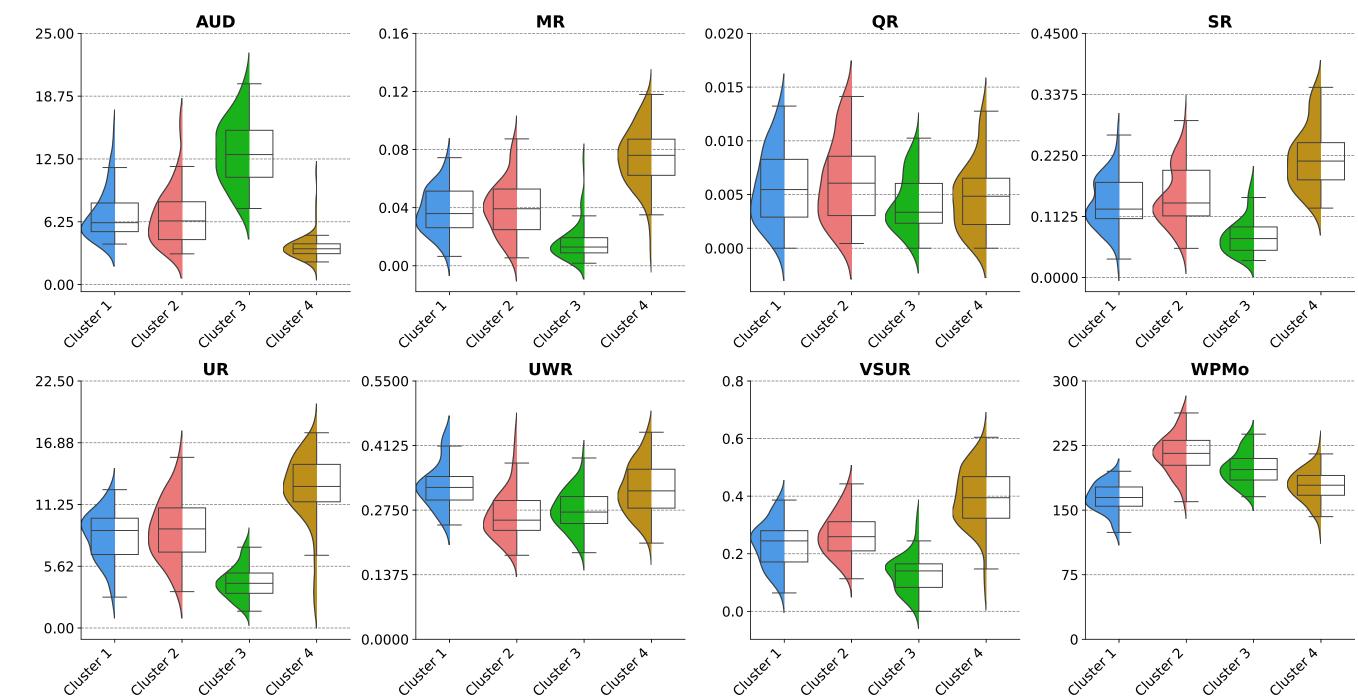

Fig. 10. Summary of profiles identified in *Public Talks* (RQ3).

identified within each type of discourse during RQ3 could be used to provide a more detailed analysis of the speakers. In this way, it would be possible to determine which expert speech profile the speaker's delivery is most aligned with.

The MLP trained and evaluated in RQ2 aimed to analyze the separability of the different discourse types. Although the model struggles with *Online Courses*, its overall performance suggests its potential for classifying speeches in open data sources. For instance, it could be used to categorize speeches on educational platforms, YouTube, or social media.However, further evaluation is needed in this area to assess the model's performance on larger datasets and its behavior when seeing new discourse types. Finally, the speech metrics proposed in this article, along with the algorithms used, could be applied to analyze other types of communicative tasks, such as interviews or debates.

5.3. Limitations

Although our dataset is comprehensive, encompassing 1250 instructional speeches, its composition remains the main limitation of our findings. Incorporating more videos from a wider range of speakers, institutions, platforms, and contexts could lead to more robust and generalizable conclusions. It is also worth noting that our findings are limited to English speeches, and the results may vary in other languages. Metrics related to speech speed are highly dependent on the language. In addition, some algorithms should be adapted to be used in other languages, such as the identification of filler words or questions. Additionally, we have not accounted for the effects of accents, the speaker's gender, or environmental conditions such as temperature.

Another limitation concerns the model explainability analysis conducted to assess the relevance of the speech metrics for discourse classification. We applied SHAP to estimate the contribution of each speech metric to the predictions of our MLP classifier. This model achieved an F1 score of 0.78, a performance attributed to the high overlap between *Online Courses* and the other discourse types. While SHAP is a robust post-hoc technique for interpreting model outputs, the resulting feature importance patterns are inherently model-dependent. Consequently, the SHAP-based insights reported in this study should be interpreted in light of the MLP's predictive performance.

The accuracy of our speech metrics also depends on the computational algorithms utilized. We did not rely on manual transcriptions; instead, we employed Whisper to generate them. Although Whisper is known for its strong performance in audio transcription, it is still an AI model that can occasionally make errors, particularly in the presence of background noise, overlapping speech, or non-standard accents. These transcription errors may introduce noise into the data and influence subsequent analyses. Although we selected high-quality audio recordings, it is possible that the quality is not optimal during all speeches, for example, lecture recordings where the speaker's microphone is not positioned close to the teacher for short periods. Finally, our analyses focused exclusively on speech metrics derived from the audio signal and its transcription.

Nonverbal behaviors, such as gestures, facial expressions, posture, and eye contact, play a significant role in communication but were not included in this study. This analysis would require complex computer vision techniques to identify behaviors, define new metrics, and filter discourses in which the speaker is not visualized, such as all the *Animated Lessons* or *Online Courses* that only display a digital blackboard and slides. Analyzing the speaker's body language could be a relevant future work, but it would not enable the use of all the instructional discourses employed in this manuscript.

6. Conclusions and future work

Oral communication plays a crucial role in instructional environments. Nevertheless, it remains unclear what characterizes expert speech, especially given the diversity of discourse types and the need to adapt speech to the particular characteristics of each. To address this problem, we conducted a comprehensive analysis of 1250 instructional videos from five types of discourses provided by experts: *Lectures*, *Online Courses*, *Animated Lessons*, *Supplementary Lessons*, and *Public Talks*. For these discourses, we extracted 16 speech metrics, and an additional six multiple-participant metrics for *Lectures*.

In the analysis of expert speech metrics, the *Lectures* exhibited more time dedicated to silences, shorter utterances, more frequent mumbles, and filler words, indicating a less refined speech. *Public Talks* contained long periods of silence, short utterances, and a high frequency of mumbles, although these were less prominent than in *Lectures*. In addition, *Public Talks* showed higher subjectivity and positive emotion. *Online Courses* did not stand out for having particularly high or low values in any of the considered speech metrics. *Animated Lessons* had the slowest speech speed and the highest proportion of unique words, combined with long utterances, few filler words, and mumbles. Considering the previous metrics, *Animated Lessons* exhibited highly refined speech. We observed a different pattern in *Supplementary Lessons*, characterized by the lowest time dedicated to pauses and long utterances.

We also analyzed the separability of the types of discourses based on the speeches. The UMAP scatter plot showed considerable separability among the majority of discourse types, except for *Online Courses*. We also developed a discourse classifier, an MLP that achieved an F1 score of 0.78. The model struggled to identify *Online Courses*, which is indicative of the high dispersion of *Online Courses* speech metrics and their overlap with other discourse types. Using explainable artificial intelligence, we observed that the MLP assigned greater importance to the ratio of unique words when classifying the discourses. Finally, we explored speech profiles in *Lectures*, *Online Courses*, and *Public Talks*, successfully identifying four, three, and four profiles respectively.

Future research could evaluate expert speech in languages other than English and compare the resulting patterns with those reported in this study. It would also be valuable to examine how speeches vary across cultural and social contexts. In addition, our work could be extended to discourse types beyond the instructional domain and to other categories of speech. For instance, debates and interviews could be considered. Alternative taxonomies could also be explored to classify instructional speech. Ultimately, future research may build upon the findings of this study to develop platforms for training and assessing instructional discourses. Such platforms could compare user speech metrics with those of expert speakers in each discourse type, thereby providing feedback that is both accessible and easy to interpret. The outcomes of our research provide practical guidelines for both speakers and researchers, intended to enhance instructional discourses.

CRediT authorship contribution statement

Mariano Albaladejo-González: Conceptualization, Methodology, Software, Validation, Formal Analysis, Data Curation, Writing – Original Draft, Visualization; **Manuel J. Gomez:** Methodology, Validation, Formal Analysis, Data Curation, Writing – Review & Editing, Visualization; **Óscar Cánovas:** Conceptualization, Methodology, Writing – Review & Editing, Supervision; **Félix Gómez Mármol:** Conceptualization, Writing – Review & Editing, Supervision, Project administration; **José A. Ruipérez-Valiente:** Conceptualization, Writing – Review & Editing, Supervision, Project administration.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: José Antonio Ruipérez Valiente reports financial support was provided by Fundación Séneca. Mariano Albaladejo González reports financial support was provided by Ministerio de Ciencia, Innovación y Universidades. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A.

Table A.4 summarizes the speech metrics of experts in instructional discourses. For each metric and discourse type, we report the three quartiles, the mean, and the standard deviation, providing precise reference values that future researchers can use to assess other speeches. MSPR, MSUR, OVR, OVUR, PEQ, and TTC metrics measure multiple-participant interactions. Therefore, they are only applicable to *Lectures*, in which student interventions may be present alongside the educator's speech.

Table A.4
Detailed description of the expert speech metrics for each type of discourse.

	Metric	*Lectures*	*Online Courses*	*Animated Lessons*	*Supplementary Lessons*	*Public Talks*
Mean	ALD	1.4628	0.9678	0.9739	0.5921	1.3746
	AUD	6.8785	9.3277	10.9161	12.9366	8.4171
	CQR	0.0023	0.0003	0.0001	0.0002	0.0002
	FLD	0.0018	0.0002	0.0000	0.0002	0.0000
	MR	0.0460	0.0332	0.0312	0.0295	0.0457
	PL	0.0987	0.1090	0.1019	0.0866	0.1292
	QR	0.0095	0.0040	0.0035	0.0050	0.0056
	SPMo	226.5308	247.0831	235.9480	241.1213	243.1757
	SPMu	299.6766	286.1154	264.6437	284.9108	276.9860
	SR	0.2149	0.1048	0.0811	0.0693	0.1560
	SJ	0.4293	0.4248	0.4501	0.3852	0.4616
	UR	9.3144	7.4897	5.8900	7.2015	8.8973
	UWR	0.1247	0.3407	0.4659	0.2076	0.3007
	VSUR	0.3308	0.1753	0.1100	0.2385	0.2678
	WPMo	178.0528	179.6498	167.6739	195.0143	188.5233
	WPMu	245.3236	212.7882	192.3861	234.0679	220.8260
	MSPR	0.9382				
	MSUR	0.9280				
	OVR	0.0019				
	OVUR	0.0215				
	PEQ	0.0825				
	TTC	0.5248				
Std	ALD	0.6886	0.5917	0.5806	0.4509	1.7740
	AUD	4.7548	7.2650	5.8106	8.8294	6.8446
	CQR	0.0030	0.0012	0.0004	0.0006	0.0005
	FLD	0.0035	0.0016	0.0001	0.0005	0.0001
	MR	0.0233	0.0236	0.0211	0.0231	0.0323
	PL	0.0388	0.0605	0.0519	0.0667	0.0477
	QR	0.0060	0.0056	0.0035	0.0040	0.0046
	SPMo	37.8088	31.8835	24.5351	24.4701	31.6964
	SPMu	44.9670	51.1702	46.9737	34.4054	42.8219
	SR	0.1057	0.0605	0.0421	0.0535	0.0841
	SJ	0.0455	0.0770	0.0621	0.0787	0.0491
	UR	3.7798	3.5478	2.4931	3.9624	4.0690
	UWR	0.0280	0.1258	0.0652	0.0552	0.0613
	VSUR	0.1243	0.1197	0.0939	0.1308	0.1319
	WPMo	30.8755	29.3922	20.3170	18.4895	27.7004
	WPMu	38.3358	43.4579	35.8057	28.8905	38.3354
	MSPR	0.1146				
	MSUR	0.1170				
	OVR	0.0030				
	OVUR	0.0303				
	PEQ	0.0912				
	TTC	0.6440				

Table A.4
continued.

	Metric	*Lectures*	*Online Courses*	*Animated Lessons*	*Supplementary Lessons*	*Public Talks*
Q1	ALD	0.9722	0.6497	0.7283	0.4184	0.8651
	AUD	3.6913	4.7719	7.0921	7.0026	4.0048
	CQR	0.0005	0.0000	0.0000	0.0000	0.0000
	FLD	0.0000	0.0000	0.0000	0.0000	0.0000
	MR	0.0282	0.0155	0.0167	0.0155	0.0208
	PL	0.0710	0.0716	0.0708	0.0388	0.0985
	QR	0.0051	0.0000	0.0013	0.0020	0.0027
	SPMo	198.7292	225.8257	221.4984	225.3778	221.0751
	SPMu	269.4623	252.1453	232.2392	263.5633	247.7965
	SR	0.1323	0.0609	0.0552	0.0352	0.0952
	SJ	0.4006	0.3799	0.4175	0.3294	0.4316
	UR	6.1395	5.0000	4.2292	4.4750	5.6586
	UWR	0.1047	0.2455	0.4403	0.1705	0.2569
	VSUR	0.2315	0.0833	0.0417	0.1389	0.1667
	WPMo	153.3470	158.3087	155.1947	184.7233	169.0119
	WPMu	222.2334	180.7235	167.7479	216.5898	192.5555
	MSPR	0.9534				
	MSUR	0.9342				
	OVR	0.0002				
	OVUR	0.0036				
	PEQ	0.0223				
	TTC	0.1250				
Q2	ALD	1.2983	0.8221	0.8367	0.4967	1.0472
	AUD	5.4437	6.6674	9.7641	10.2490	6.0085
	CQR	0.0010	0.0000	0.0000	0.0000	0.0000
	FLD	0.0004	0.0000	0.0000	0.0000	0.0000
	MR	0.0424	0.0292	0.0257	0.0257	0.0404
	PL	0.0995	0.1064	0.0998	0.0931	0.1265
	QR	0.0081	0.0019	0.0028	0.0042	0.0050
	SPMo	223.0259	247.7290	233.9729	239.1504	241.5902
	SPMu	297.3264	278.2340	251.1512	279.2987	274.6065
	SR	0.2018	0.0951	0.0704	0.0548	0.1425
	SJ	0.4312	0.4285	0.4516	0.3899	0.4603
	UR	9.0986	7.0333	5.3333	6.4143	9.0263
	UWR	0.1199	0.3259	0.4819	0.1943	0.2961
	VSUR	0.3267	0.1616	0.0893	0.2277	0.2593
	WPMo	176.6107	176.8664	165.0673	196.2051	185.8287
	WPMu	244.9249	209.9538	182.0408	232.6703	217.3280
	MSPR	0.9783				
	MSUR	0.9691				
	OVR	0.0009				
	OVUR	0.0118				
	PEQ	0.0531				
	TTC	0.3334				
Q3	ALD	1.8180	1.0591	1.0067	0.6202	1.2915
	AUD	8.2266	10.9614	13.1519	16.7061	10.5000
	CQR	0.0030	0.0000	0.0000	0.0000	0.0003
	FLD	0.0017	0.0000	0.0000	0.0000	0.0000
	MR	0.0619	0.0453	0.0406	0.0375	0.0627
	PL	0.1259	0.1415	0.1301	0.1323	0.1600
	QR	0.0127	0.0054	0.0046	0.0071	0.0077
	SPMo	254.1253	268.9187	248.2262	255.9423	264.3846
	SPMu	331.8966	315.8175	286.5377	297.5724	302.1649
	SR	0.2772	0.1372	0.0916	0.0859	0.2004
	SJ	0.4619	0.4655	0.4859	0.4463	0.4954
	UR	11.9847	9.5000	6.8333	9.1429	11.9272
	UWR	0.1407	0.4126	0.5062	0.2337	0.3322
	VSUR	0.4118	0.2495	0.1582	0.3117	0.3447
	WPMo	201.7222	198.0438	175.3928	205.6854	206.8382
	WPMu	272.3278	240.4006	211.3724	247.6527	242.8927
	MSPR	0.9912				
	MSUR	0.9846				
	OVR	0.0023				
	OVUR	0.0281				
	PEQ	0.1101				
	TTC	0.6797				

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